



QATIPv3 Azure Lab3

VNet deployment using Terraform

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Lab Objectives

1. In this lab, you will create the following resources...

Resource Group - “**rg3-{suffix}**” in West Europe

VNet - Using 10.0.0.0/16

Subnet - Using 10.0.1.0/24

Routing Table - Routing 0.0.0.0 traffic to the internet

Routing Table Association – between the subnet and the routing table

NSG - Allowing inbound RDP 3389 traffic

Security group association – between the subnet and the NSG

Virtual machine – “**VM3**” with a virtual nic and public IP

This lab breaks a deployment into phases, each a separate task, to demonstrate component interdependencies:

Task 1: Resource Group, VNet and subnet

Task 2: Routing Table and Routing Table Association

Task 3: Network Security Group and Network Security Group association

Task 4: Virtual Machine with Virtual Nic and Public IP

Task 5: Testing

Teaching Points

This lab will take you through coding a multi-component cloud deployment into Azure using Terraform. You will deploy a virtual machine into a relatively simple network but even this has several components that need either to exist already or be created as part of the deployment. You have the option of attempting the tasks yourself with minimal guidance or following step-by-step instructions.

Before you begin

1. Ensure you have completed Lab0 before attempting this lab
2. In the IDE terminal pane, enter the following command... `cd c:\azurelabs\lab3`
3. This shifts your current working directory to lab3. Ensure all commands are executed in this directory
4. Close any open files and use the Explorer pane to navigate to and open the lab3 folder

Solution

The solution to this lab is in folder C:\azurelab\solutions\lab3. Try to use this only as a last resort if you are struggling to complete the step-by-step processes.

Task 1. Create the Resource group, VNet and Subnet

Try it yourself

Searching the Registry for guidance....

1. Add and configure an **azurerm_provider** block into main.tf using version **4.16.0** and referencing your lab subscription.
2. Create an **azurerm_resource_group** resource group named **rg3-{suffix}**. Specify **West Europe** as the location.
3. Create an example **azurerm_virtual_network** resource within the resource group. Do not provision inline subnets or DNS Servers. Use a **10.0.0.0/16** address space and reference the name and location of the **azurerm_resource_group** you created in step 2.
4. Create a single example **azurerm_subnet** within the virtual network. Delegation is not required. Use a **10.0.1.0/24** address space and reference the **azurerm_resource_group** and **azurerm_virtual_network** resources created in steps 2 and 3.
5. Deploy these resources and validate their creation through the Azure Portal

Step-by-Step

1. Open Terraform Registry Resources link [here](#)
2. Scroll down to the 'Example Usage' section and copy the sample code into the Task1 section of main.tf
3. Change the version value to **4.16.0**
4. Remove the **resource_provider_registrations** line
5. Add the following line under 'features {}', replacing <your subscription id here> with your Azure lab subscription id..
subscription_id = "<your subscription id here>"
6. Change the name of the resource group to **rg3-{suffix}**
7. Note how the **azurerm_virtual_network** resource block has implicit dependency on the **azurerm_resource_group**..

```
resource_group_name = azurerm_resource_group.example.name  
location = azurerm_resource_group.example.location
```

8. We will now add a subnet to the network. Open the Terraform Registry Resources link [here](#)
9. Scroll down to the 'Example Usage' section and copy **just** the 'azurerm_subnet' example block into main.tf, below your existing Task1 code.
10. Note how the new resource has implicit dependencies on both the azurerm_resource_group and azurerm_virtual_network...

```
resource_group_name = azurerm_resource_group.example.name  
virtual_network_name = azurerm_virtual_network.example.name
```

11. Delete the entire **delegation** sub-block
12. Initialize terraform with **terraform init**
13. Deploy these resources using **terraform plan** followed by **terraform apply**. Enter **yes** when prompted.
14. Switch to the Azure Portal. Verify the creation of your resource group, the vnet with a single subnet.

Task 2. Create Routing Table and Routing Table Association

Try it yourself

1. Create an **azurerm_route_table** resource. Reference it to your azurerm_resource_group name and location. Configure local routing for all 10.0.1.0/16 destinations and the Internet for all others.
2. Create an **azurerm_subnet_route_table_association** resource. Associate it with the **azurerm_subnet** and **azurerm_route_table** resources you created in Task 1.
3. Deploy these resources and validate their creation through the Azure Portal

Step-by-Step

1. Open Terraform Registry Resources link [here](#)
2. Familiarize yourself and then copy the **azurerm_route_table** example block into the Task2 section of main.tf **excluding the resource group resource block**.

3. Note how the new resource has implicit dependencies on `azurerm_resource_group...`
location = azurerm_resource_group.example.location
resource_group_name = azurerm_resource_group.example.name
4. Change the `address_prefix` to **"10.0.0.0/16"** so that it aligns to the vnet address space defines in Task 1.
5. Remove the **tags** sub-block
6. Add a second route block to add a default route to the internet...

```
route {  
  name = "Route-Internet"  
  address_prefix = "0.0.0.0/0"  
  next_hop_type = "Internet"  
}
```
7. We now need to associate our subnet with the route table. Open Terraform Registry Resources link [here](#)
8. Copy the **last** resource block example, `'azurerm_subnet_route_table_association'`, into the bottom of the Task 2 block of `main.tf`
9. Note how the new resource has implicit dependencies on `azurerm_subnet` and `azurerm_route_table...`
subnet_id = azurerm_subnet.example.id
route_table_id = azurerm_route_table.example.id
10. Deploy these resources using **terraform plan** followed by **terraform apply**
11. Switch to the Azure Portal. Verify the creation of 2 additional resources within your resource group...

Routes			
Search routes			
Name	↕	Address prefix	↕ Next hop type
Route-Internet		0.0.0.0/0	Internet
route1		10.0.0.0/16	Virtual network
Subnets			
Search subnets			
Name	↕	Address range	↕ Virtual network
example-subnet		10.0.1.0/24	example-network

Task 3. Create Network Security Group

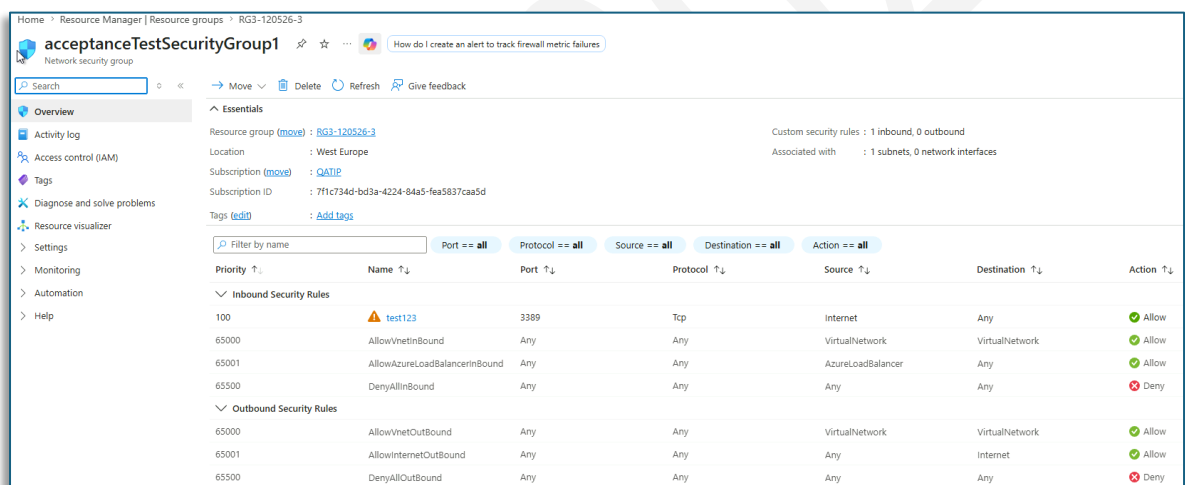
Try it yourself

1. Create an **azurerm_network_security_group** resource. Reference it to your resource group name and location. Configure the security rule to permit inbound traffic from the internet trying to reach any destination on TCP port 3389.
2. Associate this NSG with your Vnet subnet using an **azurerm_subnet_network_security_group_association** resource.
3. Deploy these resources and validate their creation through the Azure Portal.

Step-by-Step

1. Open Terraform Registry Resources link [here](#)
2. Familiarize yourself and then copy the **azurerm_network_security_group** example block into Task3 section of main.tf
3. Note how the new resource has implicit dependencies on azurerm_resource_group...
location = azurerm_resource_group.example.location
resource_group_name = azurerm_resource_group.example.name
4. Change the destination_port_range to **"3389"**
5. Change the source_address_prefix to **"Internet"**
6. Remove the **tags** sub-block
7. We must now associate this security group with our subnet. Open Terraform Registry Resources link [here](#)

8. Familiarize yourself and then copy the `'azurerm_subnet_network_security_group_association'` sample block into the bottom of the Task 3 block in `main.tf`
9. Note how the new resource has implicit dependencies on `azurerm_subnet` and `azurerm_network_security_group`...
`subnet_id = azurerm_subnet.example.id`
`network_security_group_id = azurerm_network_security_group.example.id`
10. Deploy these 2 new resources using **terraform apply** followed by **yes**
11. Switch to the Azure Portal. Click into your security group to verify the security rule and subnet association are created... The warning icon is highlighting the fact that a rule allows inbound traffic from any external IP. In production you would create more restrictive rules.



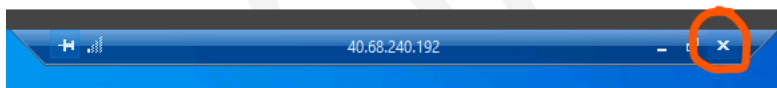
Task 4. Create Virtual Machine with Virtual Nic and Public IP

1. The code to create a virtual machine, virtual nic and a static public ip has been pre-provisioned for lab brevity.
2. In `main.tf`, comment out the line with `"*/"` at the beginning, and the line with `"\*"` at the end of the Task 4 code block. This uncomments the entire block.
3. **Note.** Address any resource reference errors if your resource identifiers do not match those expected.
4. Deploy these 3 new resources using **terraform apply**

5. **Note.** If you receive errors regarding the machine size being used ("Standard_B2s") then try "Standard_D2s_V3", "Standard_DS2_V2" or "Standard_DS3_v2".

Task 5. Testing

1. Return to the Azure Portal
2. Review the list of resources in your resource group and then click on virtual machine **VM3**
3. From the top menu click **Connect > Connect > Download RDP**
4. Download the RDP file, as prompted, and open
5. Connect & Accept the certificate validation and when prompted, login with your credentials from the terraform file (VM configuration) **Note:** the virtual machine may take a little while to become ready to accept an RDP connection
6. Log in using **TFadminuser/TFP@\$\$w0rd1234!**
7. Having verified connectivity, you can disconnect from the virtual machine without waiting for it to fully initialize...



Task 6. Lab Clean-up

1. Run **terraform destroy** and confirm with **yes** when prompted.

****** Congratulations, you have completed this lab ******

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